25. a. A linear transformation of vector spaces is analogous to a homomorphism of groups.

b. The **kernel** or **nullspace** of ϕ is $\text{Ker}(\phi) = \phi^{-1}[\{0'\}] = \{\alpha \in V \mid \phi(\alpha) = 0'\}$. Considering just the additive groups of V and V', group theory shows that $\text{Ker}(\phi)$ is an additive group. Let $\alpha \in \text{Ker}(\phi)$. Then $\phi(a\alpha) = a\phi(\alpha) = a0' = 0'$, so $\text{Ker}(\phi)$ is closed under scalar multiplication by scalars $a \in F$. Hence $\text{Ker}(\phi)$ is a subspace of V.

c. ϕ is an isomorphism of V with V' if ϕ is one to one (equivalently, if $\text{Ker}(\phi) = \{0\}$) and if ϕ maps V onto V'.

26. Let V/S be the factor group $\langle V, +\rangle/\langle S, +\rangle$, which is abelian because V is abelian. Define scalar multiplication on V/S by $a(\alpha + S) = a\alpha + S$ for $a \in F, (\alpha + S) \in V/S$. For $\sigma \in S$, we have $a(\alpha + \sigma) = a\alpha + a\sigma$ and $a\sigma \in S$ for all σ in the subspace S, so this scalar multiplication is well defined, independent of the choice of representative in the coset $\alpha + S$. Because addition and scalar multiplication in V/S are computed in terms of representatives in V and because V is a vector space, we see that addition and scalar multiplication in V/S satisfy the axioms for a vector space.

27. a. By group theory, we know that $\langle \phi[V], +\rangle$ is a subgroup of $\langle V', +\rangle$. Let $\alpha \in V$ and $a \in F$. Then $a\phi(\alpha) = \phi(a\alpha)$ shows that $\phi[V]$ is closed under multiplication by scalars in F. Thus $\phi[V]$ is a subspace of V'.

b. Let $\{\alpha_1, \alpha_2, \cdots, \alpha_r\}$ be a basis for $\text{Ker}(\phi)$. By Theorem 30.19, this set can be enlarged to a basis

$$\{\alpha_1, \alpha_2, \cdots, \alpha_r\, \beta_1, \beta_2, \cdots, \beta_m\}$$

for V. Let $\gamma \in V$. Then $\gamma = a_1\alpha_1 + \cdots + a_r\alpha_r + b_1\beta_1 + \cdots + b_m\beta_m$ for scalars $a_i, b_j \in F$. Because $\phi(\alpha_i) = 0$ for $i = 1, \cdots, r$, we see that $\phi(\gamma) = b_1\phi(\beta_1) + \cdots + b_m\phi(\beta_m)$. Thus $\{\phi(\beta_1), \cdots, \phi(\beta_m)\}$ spans $\phi[V]$. We claim that this set is independent, and hence is actually a basis for $\phi[V]$. Suppose that $c_1\phi(\beta_1) + \cdots + c_m\phi(\beta_m) = 0$ for scalars $c_j \in F$. Then $\phi(c_1\beta_1 + \cdots + c_m\beta_m) = 0$, so $(c_1\beta_1 + \cdots + c_m\beta_m) \in$ $\text{Ker}(\phi)$, and thus

$$c_1\beta_1 + \cdots + c_m\beta_m = d_1\alpha_1 + \cdots + d_r\alpha_r$$

for some scalars d_i. Moving everything to the lefthand side of this equation, we obtain a linear combination of the vectors α_i and β_j which is equal to 0. Because the α_i and β_j form a basis for V, they are independent so all the coefficients d_i and c_j must be zero. The fact that the c_j must be zero shows that $\{\phi(\beta_1), \cdots, \phi(\beta_m)\}$ is independent, and thus is a basis for $\phi[V]$. By our construction, $\dim(\text{Ker}(\phi)) = r$, $\dim(V) = r + m$, and we have shown that $\dim(\phi[V]) = m$. Thus $\dim(\phi[V]) = m = (r + m) - r = \dim(V) - \dim(\text{Ker}(\phi))$.

31. Algebraic Extensions

1. Because $\text{irr}(\sqrt{2}, \mathbb{Q}) = x^2 - 2$, the degree is 2 and a basis is $\{1, \sqrt{2}\}$.

2. By Example 31.9, the degree is 4 and a basis is $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$.

3. We notice that $\sqrt{18} = \sqrt{2}\sqrt{3}\sqrt{3}$. Thus $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{18})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ are the same field. Thus the degree is 4 and a basis is $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$ by Example 31.9.

4. Now $\sqrt{3} \notin \mathbb{Q}(\sqrt[3]{2})$ because $\mathbb{Q}(\sqrt{3})$ is of degree 2 over $\mathbb{Q}$ while $\mathbb{Q}(\sqrt[3]{2})$ is of degree 3, and 2 does not divide 3. Thus the degree of $\mathbb{Q}(\sqrt[3]{2}, \sqrt{3})$ over $\mathbb{Q}$ is 6. We form products from the bases $\{1, \sqrt{3}\}$ for $\mathbb{Q}(\sqrt{3})$ over $\mathbb{Q}$ and $\{1, \sqrt[3]{2}, (\sqrt[3]{2})^2\}$ for $\mathbb{Q}(\sqrt[3]{2})$ over $\mathbb{Q}(\sqrt{3})$, obtaining $\{1, \sqrt[3]{2}, (\sqrt[3]{2})^2, \sqrt{3}, \sqrt{3}(\sqrt[3]{2}), \sqrt{3}(\sqrt[3]{2})^2\}$ as a basis.

5. As in the solution to Exercise 4, the extension has degree 6. Taking products from bases $\{1, \sqrt{2}\}$ for $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$ and $\{1, \sqrt[3]{2}, (\sqrt[3]{2})^2\}$ for $\mathbb{Q}(\sqrt[3]{2})$ over $\mathbb{Q}(\sqrt{2})$, we see $\{1, \sqrt[3]{2}, (\sqrt[3]{2})^2, \sqrt{2}, \sqrt{2}(\sqrt[3]{2}), \sqrt{2}(\sqrt[3]{2})^2\}$ is a basis. It is easy to see that $\mathbb{Q}(\sqrt{2}, \sqrt[3]{2}) = \mathbb{Q}(\sqrt[6]{2})$ since $2^{1/6} = 2^{7/6}/2 = 2^{3/6}2^{4/6}/2 = 2^{1/2}(2^{1/3})^2/2$, so another basis is $\{1, 2^{1/6}, 2^{2/6}, 2^{3/6}, 2^{4/6}, 2^{5/6}\}$.

6. As shown in Example 31.9, we have $\deg(\sqrt{2} + \sqrt{3}, \mathbb{Q}) = 4$, so $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ and a basis over $\mathbb{Q}$ is $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$ just as in Example 31.9.

7. Because $\sqrt{2}\sqrt{3} = \sqrt{6}$, we see that the field is $\mathbb{Q}(\sqrt{6})$ which has degree 2 over $\mathbb{Q}$ and a basis $\{1, \sqrt{6}\}$.

8. As in the solution to Exercise 4, we see the extension is of degree 6 because 2 does not divide 3. We form products from the bases $\{1, \sqrt{2}\}$ for $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$ and $\{1, \sqrt[3]{5}, (\sqrt[3]{5})^2\}$ for $\mathbb{Q}(\sqrt[3]{5})$ over $\mathbb{Q}(\sqrt{2})$, yielding $\{1, \sqrt[3]{5}, (\sqrt[3]{5})^2, \sqrt{2}, \sqrt{2}(\sqrt[3]{5}), \sqrt{2}(\sqrt[3]{5})^2\}$ as a basis.

9. Now $\sqrt[3]{6}/\sqrt[3]{2} = \sqrt[3]{3}$ and $\sqrt[3]{24} = 2(\sqrt[3]{3})$, so $\mathbb{Q}(\sqrt{2}, \sqrt[3]{6}, \sqrt[3]{24}) = \mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{3})$. The degree over $\mathbb{Q}$ is 9, and we take products from the bases $\{1, \sqrt[3]{2}, (\sqrt[3]{2})^2\}$ and $\{1, \sqrt[3]{3}, (\sqrt[3]{3})^2\}$ for $\mathbb{Q}(\sqrt[3]{2})$ over $\mathbb{Q}$ and $\mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{3})$ over $\mathbb{Q}(\sqrt[3]{2})$ respectively, obtaining the basis

$$\{1, \sqrt[3]{2}, \sqrt[3]{4}, \sqrt[3]{3}, \sqrt[3]{6}, \sqrt[3]{12}, \sqrt[3]{9}, \sqrt[3]{18}, \sqrt[3]{36}\}.$$

10. Because $\mathbb{Q}(\sqrt{2}, \sqrt{6}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, the extension is of degree 2 over $\mathbb{Q}(\sqrt{3})$ and we can take the set $\{1, \sqrt{2}\}$ as a basis.

11. Example 31.9 shows that $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, so the extension has degree 2 and we can take as basis over $\mathbb{Q}(\sqrt{3})$ the set $\{1, \sqrt{2}\}$.

12. By Example 31.9, $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ so the degree of the extension is 1 and $\{1\}$ is a basis.

13. Now $\sqrt{6} + \sqrt{10} = \sqrt{2}(\sqrt{3} + \sqrt{5})$ so we have $\mathbb{Q}(\sqrt{2}, \sqrt{6} + \sqrt{10}) = \mathbb{Q}(\sqrt{2}, \sqrt{3} + \sqrt{5})$. The degree of the extension is 2 and a basis over $\mathbb{Q}(\sqrt{3} + \sqrt{5})$ is $\{1, \sqrt{2}\}$.

14. The definition is incorrect. An algebraic extension need not be a finite extension.

An **algebraic extension** of a field F is an extension field E of F with the property that each $\alpha \in E$ is a zero of some nonzero polynomial in $F[x]$.

15. The definition is incorrect. If an element adjoined is transcendental over F, the extension is not finite.

A **finite extension field** of a field F is an extension field of F that has finite dimension when regarded as a vector space having F as its field of scalars.

16. The definition is correct.

17. The definition is not quite correct. We should insert "nonconstant" before "polynomial" and "in F[x]" after it.

A field F is **algebraically closed** if and only if every nonconstant polnomial in $F[x]$ has a zero in F.

18. Let $E = \mathbb{Q}(\sqrt{2})$ and let $F = \mathbb{Q}$. The algebraic closure of $\mathbb{Q}$ in $\mathbb{Q}(\sqrt{2})$ is $\mathbb{Q}(\sqrt{2})$ because it is an algebraic extension of $\mathbb{Q}$. However, $\mathbb{Q}(\sqrt{2})$ is not algebraically closed, because the polynomial $x^2 + 1$ has no zeros in $\mathbb{Q}(\sqrt{2})$.

19. F T F T F T F F F F

20. Because E is a finite dimensional vector space over F, the set of all powers of an element α in E cannot be independent, so some finite linear combination with nonzero coefficents in F of these powers must be zero.

21. Taking a basis of n elements α_i for E over F and a basis of m elements β_j for K over E, the set of the mn possible products $\alpha_i\beta_j$ is a basis for K over F.

22. If $b \neq 0$, then $a + bi \in \mathbb{C}$ but $a + bi \notin \mathbb{R}$. By Theorem 31.3, $a + bi$ is algebraic over $\mathbb{R}$. Then by Theorem 31.4,

$$[\mathbb{C} : \mathbb{R}] = [\mathbb{C} : \mathbb{R}(a + bi)][\mathbb{R}(a + bi) : \mathbb{R}] = 2,$$

and because $a + bi \notin \mathbb{R}$, we must have $[\mathbb{R}(a + bi) : \mathbb{R}] = 2$, so $[\mathbb{C} : \mathbb{R}(a + bi)] = 1$. Thus $\mathbb{C} = \mathbb{R}(a + bi)$.

23. Let α be any element in E that is not in F. Then $[E : F] = [E : F(\alpha)][F(\alpha) : F] = p$ for some prime p by Theorem 31.4. Because α is not in F, we know that $[F(\alpha) : F] > 1$, so we must have $[F(\alpha) : F] = p$ and therefore $[E : F(\alpha)] = 1$. As we remarked after Definition 31.2, this shows that $E = F(\alpha)$, which is what we wish to show.

24. If $x^2 - 3$ were reducible over $\mathbb{Q}(\sqrt[3]{2})$, then it would factor into linear factors over $\mathbb{Q}(\sqrt[3]{2})$, so $\sqrt{3}$ would lie in the field $\mathbb{Q}(\sqrt[3]{2})$, and we would have $\mathbb{Q}(\sqrt{3}) \leq \mathbb{Q}(\sqrt[3]{2})$. But then by Theorem 31.4,

$$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}(\sqrt{3}][\mathbb{Q}(\sqrt{3}) : \mathbb{Q}].$$

This equation is impossible because $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$ while $[\mathbb{Q}(\sqrt{3}) : \mathbb{Q}] = 2$.

25. Corollary 31.6 shows that the degree of an extension of F by successively adjoining square roots must be 2^n for some $n \in \mathbb{Z}^+$. Because $x^{14} - 3x^2 + 12$ is irreducible over $\mathbb{Q}$ by the Eisenstein condition with $p = 3$, and because $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 14$ for any zero α of this polynomial, and because 14 is not a divisor of 2^n for any $n \in \mathbb{Z}^+$, we see that α cannot lie in any field obtained by adjoining just square roots. Therefore α cannot be expressed as a rational function of square roots, square roots of rational functions of square roots, etc.

26. We need only show that for each $\alpha \in D, \alpha \neq 0$, its multiplicative inverse $1/\alpha$ is in D also. Because E is a finite extension of F, we know that α is algebraic over F. Let $\deg(\alpha, F) = n$. Then by Theorem 30.23, we have

$$F(\alpha) = \{a_0 + a_1\alpha + a_2\alpha^2 + \cdots + a_{n-1}\alpha^{n-1} \mid a_i \in F \text{ for } i = 0, \cdots n - 1\}.$$

In particular, $1/\alpha \in F(\alpha)$, so $1/\alpha$ is equal to polynomial in α with coefficients in F, and is in D.

27. Obviously $\mathbb{Q}(\sqrt{3} + \sqrt{7}) \subseteq \mathbb{Q}(\sqrt{3}, \sqrt{7})$. Now $(\sqrt{3} + \sqrt{7})^2 = 10 + 2\sqrt{21}$ so $\sqrt{21} \in \mathbb{Q}(\sqrt{3} + \sqrt{7})$. Hence $(3\sqrt{7} + 7\sqrt{3}) - 7(\sqrt{3} + \sqrt{7}) = -4\sqrt{7}$ is in $\mathbb{Q}(\sqrt{3} + \sqrt{7})$, so this field contains $\sqrt{7}$ and also $(\sqrt{3} + \sqrt{7}) - \sqrt{7} = \sqrt{3}$. Therefore $\mathbb{Q}(\sqrt{3}, \sqrt{7}) \subseteq \mathbb{Q}(\sqrt{3} + \sqrt{7})$, so $\mathbb{Q}(\sqrt{3}, \sqrt{7}) = \mathbb{Q}(\sqrt{3} + \sqrt{7})$. [One can also make an argument like that in Example 31.9 of the text, finding $\text{irr}(\sqrt{3} + \sqrt{7}, \mathbb{Q})$ and showing that it is of degree 4 over $\mathbb{Q}$. Then we would have $[\mathbb{Q}(\sqrt{3}, \sqrt{7}) : \mathbb{Q}(\sqrt{3} + \sqrt{7})] = 1$, so the fields are equal.]

28. If $a = b$ the result is clear; we assume $a \neq b$. It is obvious that $\mathbb{Q}(\sqrt{a} + \sqrt{b}) \subseteq \mathbb{Q}(\sqrt{a}, \sqrt{b})$.

We now show that $\mathbb{Q}(\sqrt{a}, \sqrt{b}) \subseteq \mathbb{Q}(\sqrt{a} + \sqrt{b})$. Let $\alpha = \frac{a-b}{\sqrt{a}+\sqrt{b}} \in \mathbb{Q}(\sqrt{a} + \sqrt{b})$. Now $\alpha = \sqrt{a} - \sqrt{b}$. Thus $\mathbb{Q}(\sqrt{a} + \sqrt{b})$ contains $\frac{1}{2}[\alpha + (\sqrt{a} + \sqrt{b})] = \frac{1}{2}(2\sqrt{a}) = \sqrt{a}$ and hence also contains $(\sqrt{a} + \sqrt{b}) - \sqrt{a} = \sqrt{b}$. Thus $\mathbb{Q}(\sqrt{a}, \sqrt{b}) \subseteq \mathbb{Q}(\sqrt{a} + \sqrt{b})$.

29. If a zero α of $p(x)$ were in E, then because $p(x)$ is irreducible over F, we would have $[F(\alpha) : F] = \deg(p(x))$, and $[F(\alpha : F]$ would be a divisor of $[E : F]$ by Theorem 31.4. By hypothesis, this is not the case. Therefore $p(x)$ has no zeros in E.

30. Because $F(\alpha)$ is a finite extension of F and $\alpha^2 \in F(\alpha)$, Theorem 31.3 shows that α^2 is algebraic over F. If $F(\alpha^2) \neq F(\alpha)$, then $F(\alpha)$ must be an extension of $F(\alpha^2)$ of degree 2, because α is a zero of $x^2 - \alpha^2$. By Theorem 31.4, we would then have $2 = [F(\alpha) : F(\alpha^2)]$ dividing $[F(\alpha) : F]$, which is impossible because $[F(\alpha) : F]$ is an odd number. Therefore $F(\alpha^2) = F(\alpha)$, so $\deg(\text{irr}(\alpha^2, F)) = \deg(\text{irr}(\alpha, F)) = [F(\alpha) : F]$ which is an odd number.

31. Suppose K is algebraic over F. Then every element of K is a zero of a nonzero polynomial in F[x], and hence in E[x]. This shows that K is algebraic over E. Of course E is algebraic over F, because each element of E is also an element of K.

Conversely, suppose that K is algebraic over E and that E is algebraic over F. Let $\alpha \in K$. We must show that α is algebraic over F. Because K is algebraic over E, α is a zero of some polynomial $a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ in $E[x]$. Because E is algebraic over F, the a_i are algebraic over F for $i = 0, 1, 2, \cdots, n$. Hence $F(a_0, a_1, a_2, \cdots, a_n)$ is an extension of F of some finite degree m by Theorem 31.11. Because α is algebraic over E of degree $r \leq n$, Theorem 31.4 shows that $F(a_0, a_1, a_2, \cdots, a_n, \alpha)$ is a finite extension of F of degree $\leq mr$. By Theorem 31.3, α is algebraic over F.

32. If α is algebraic over $\overline{F}_E$, then $\overline{F}_E(\alpha)$ is algebraic over $\overline{F}_E$ and by definition, $\overline{F}_E$ is algebraic over F. By Exercise 31, then $\overline{F}_E(\alpha)$ is algebraic over F so, in particular, α is algebraic over F. But then $\alpha \in \overline{F}_E$ contrary to hypothesis. Thus α is transcendental over $\overline{F}_E$.

33. Let $f(x)$ be a nonconstant polynomial in $\overline{F}_E[x]$. We must show that $f(x)$ has a zero in $\overline{F}_E$. Now $f(x) \in E[x]$ and E is agebraically closed by hypothesis, so $f(x)$ has a zero α in E. By Exercise 32, if α is not in $\overline{F}_E$, then α is transcendental over $\overline{F}_E$. But by construction, α is a zero of $f(x) \in \overline{F}_E[x]$, so this is impossible. Hence $\alpha \in \overline{F}_E$, which shows that $\overline{F}_E$ is algebraically closed.

34. Let $\alpha \in E$ and let $p(x) = \text{irr}(\alpha, F)$ have degree n. Now $p(x)$ factors into $(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n)$ in $\overline{F}[x]$. Because by hypotheses all zeros of $p(x)$ in $\overline{F}$ are also in E, we see that this same factorization is also valid in $E[x]$. Hence

$$p(\alpha) = (\alpha - \alpha_1)(\alpha - \alpha_2) \cdots (\alpha - \alpha_n) = 0,$$

so $\alpha = \alpha_i$ for some i. This shows that $F \leq E \leq \overline{F}$. Because by definition $\overline{F}$ contains only elements that are algebraic over F and E contains all of these, we see that $E = \overline{F}$ and is therefore algebraically closed.

35. If F is a finite field of odd characteristic, then $1 \neq -1$ in F. Because $1^2 = (-1)^2 = 1$, the squares of elements of F can run through at most $|F| - 1$ elements of F, so there is some $a \in F$ that is not a square. The polynomial $x^2 - a$ then has no zeros in F, so F is not algebraically closed.

36. For all $n \in \mathbb{Z}, n \geq 2$, the polynomial $x^n - 2$ is irreducible in $\mathbb{Q}[x]$ by the Eisenstein condition with $p = 2$. This shows that $\mathbb{Q}$ has finite extensions contained in $\mathbb{C}$ of arbitrarily high degree. If $\overline{\mathbb{Q}}_{\mathbb{C}}$ were a finite extension of $\mathbb{Q}$ of degree r, then there would be no algebraic extensions of $\mathbb{Q}$ in $\mathbb{C}$ of degree greater than r. Thus the algebraic closure $\overline{\mathbb{Q}}_{\mathbb{C}}$ of $\mathbb{Q}$ in $\mathbb{C}$ cannot be a finite extension of $\mathbb{Q}$.

37. Because $[\mathbb{C} : \mathbb{R}] = 2$ and $\mathbb{C}$ is an algebraic closure of $\mathbb{R}$, it must be that every irreducible $p(x)$ in $\mathbb{R}[x]$ of degree > 1 is actually of degree 2. Because $\mathbb{C} = \mathbb{R}$ where $\alpha \in \mathbb{C}$ is any zero of any such polynomial,